Georgia Tech PHYS 6268

Nonlinear Dynamics and Chaos

Instructor: Predrag Cvitanović
Fall semester 2025

Midterm

9:30-10:45, October 14, 2025

Advice: read through the problems first, and then pick and chose which points to do first. If you get two thirds of the required 31 points, that is 100% score. All points beyond this count as bonus points.

== you have 1 1/4 hour to complete the exam
== open book, open lecture notes, you can use a calculator / laptop / tablet
== bring your own letter-size paper
== waste no time googling
== sail solo - no collaborations/discussions
== put your name on the top, and number the pages
== if doing the exam on a tablet, you have extra 10 min for Canvas upload

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Problem 1: Bifurcations in one dimension

Consider the 1D dynamical system

$$\dot{x} = 5 - re^{-x^2}$$

as a function of the parameter r.

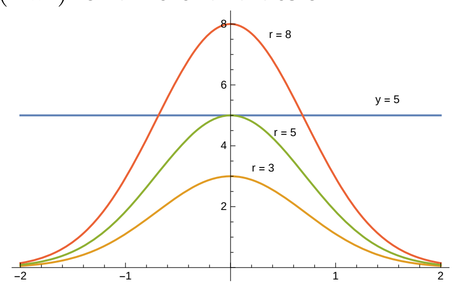
1. Find the values of r at which bifurcations occur $\boxed{\text{2 point}}$
2. Classify the bifurcations. Saddle-node, transcritical, supercritical, subcritical pitchfork? $\boxed{\text{2 point}}$
3. Sketch the bifurcation diagram of equilibria as the function of r. $\boxed{\text{1 point}}$

Solution 1: Bifurcations in one dimension

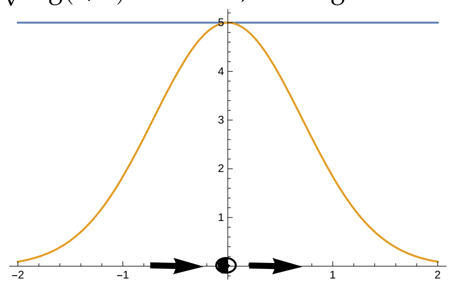
1) We can find the equilibria for $\dot{x} = 5 - r\exp(-x^2)$ by setting $\dot{x} = 0$ and solving for x:

$$x_\pm = \pm\sqrt{\log(r/5)}$$

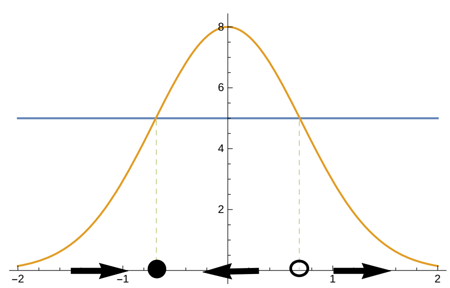
Alternatively, we can look at the bifurcation sequence by looking at the plot of y = 5 and $y = r\exp(-x^2)$ for different values of r:



Using either of these approaches we find that for r $<$ 5, there are no equilibria, at r =5 there is a single equilibrium at x = 0, and for r $>$ 5 there are two equilibria at $x = \pm\sqrt{\log(r/5)}$. For r = 5, the origin is a bistable node



For r $>$ 5, we see that the bistable node splits into a stable node and an unstable node



We confirm this by linear stability analysis, which shows that for $r = 8$ the equilibrium stability $f'(x_{\pm}) = \pm 6.02954$ is less than zero at the negative root and greater than zero at the positive root.

2) Since we have a bifurcation that generates a pair of equilibria "out of the blue," where one is stable and the other is unstable, this is a saddle-node bifurcation.

To show this more precisely, look at the Taylor series of the v(x) about the bifurcation point $(x*, r_c) = (0, 5)$:

$$v(x) \simeq \left(-(r-5) + O[r-5]^3\right) + \left(5 + (r-5) + O[r-5]^3\right) x^2 + O[x]^3$$

To the lowest order in $x - 0$ and $r - 5$, we get

$$v(x) \simeq -(r-5) + 5\,x^2\,.$$

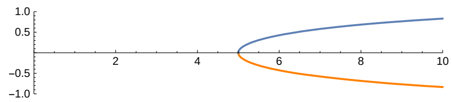
It is obvious that by the shifts and rescalings $x \to -x/\sqrt{5}$, $r - 5 \to r$, and $t \to \sqrt{5}t$, this can be reduced to the normal form for a saddle-node bifurcation

$$\frac{\mathrm{dx}}{\mathrm{dt}} = r - x^2\,.$$

3) To get a bifurcation diagram, solve for r_c as a function of x*,

$$r_c = 5\,e^{x^2}\,,$$

and then plot r_c vs x* with the axes reversed.



The upper (blue) branch corresponds to the unstable equilibrium; the lower (orange) one to the stable equilibrium.

Problem 2: A nonuniform oscillator

Consider the flow on the circle

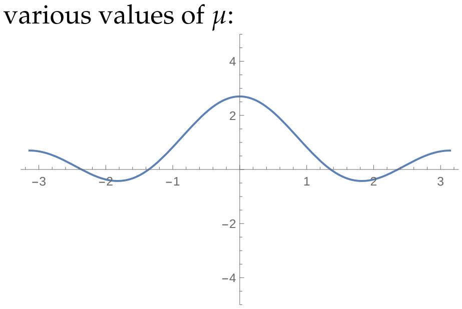
$$\dot{\theta} = \mu + \cos\theta + \cos 2\theta$$

as a function of the parameter μ.

1. Sketch the phase portrait as the function of the control parameter μ. $\boxed{\text{1 point}}$
2. Classify the bifurcations that occur as μ varies $\boxed{\text{2 point}}$
3. Find the values of μ at which bifurcations occur $\boxed{\text{2 point}}$

Solution 2: A nonuniform oscillator

1) It is obvious by inspection that changing μ just shifts the fixed curve $\cos\theta + \cos 2\theta$ up and down on the $\dot{\theta}$ vs θ plane. This can be confirmed graphically by plotting $\dot{\theta}$ vs. θ for various values of μ:



2) We expect saddle-node bifurcation when the local maxima and minima of $\dot{\theta}$ cross the horizontal axis. In order to find these, we look at the "matrix of velocity gradients," i.e., the first derivative of $\dot{\theta}$ with respect to θ:

$$\frac{\partial}{\partial\theta}\dot{\theta} = -\sin(\theta) - 2\sin(2\theta)$$

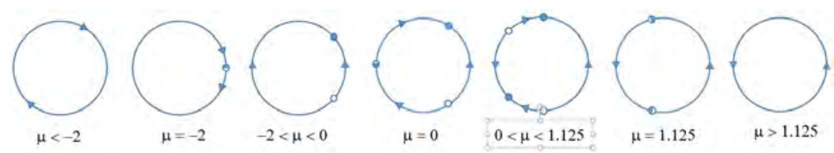
This has two trivial roots at 0 and $\pm\pi$, which correspond to the local maxima of $\dot{\theta}$. To find the other two roots, we use the trigonometric identity $\sin 2\theta = 2\sin\theta\cos\theta$ to rewrite $\partial\dot{\theta}/\partial\theta$ as

$$\begin{aligned}\frac{\partial\dot{\theta}}{\partial\theta} &= -\sin\theta - 2\sin 2\theta \\ &= -\sin\theta - 4\sin\theta\cos\theta \\ &= -\sin\theta(1 + 4\cos\theta)\end{aligned}$$

so two additional roots exist at $\theta = \cos^{-1}$(-1/4).

3) We can find the critical values of μ, by evaluating $\dot{\theta}$ at these points and setting it to zero, which gives the values of μ where the peaks and troughs just touch

the horizontal axis. As μ is increased from $-\infty$, the first bifurcation occurs at $\mu = -2$ where a pair of fixed points are created in a saddle-node bifurcation when the peak at $\theta = 0$ first crosses the horizontal axis. Below this bifurcation, there are no fixed points and the system just rotates (non-uniformly) in the clockwise direction. These separate for increasing μ, as shown below, until at $\mu = 0$ the system undergoes a second saddle-node bifurcation as the peak at $\theta = \pm\pi$ crosses the horizontal axis. As μ is increased further, the two pairs of fixed points approach each other, until at $\mu = 9/8 = 1.125$, they annihilate in simultaneous saddle-node bifurcations. For $\mu > 1.125$, there are no fixed points and the system just rotates, non-uniformly in the counter clockwise direction. This bifurcation sequence can be represented in the following series of phase portraits:



Problem 3: Analysis of a 2-dimensional flow

Consider the 2-d flow

$$\dot{x} = x(3 - y - 2x^2)\,, \qquad \dot{y} = (y-1)(x - 1/2)\,. \tag{1}$$

1. **[easy]** Determine all equilibria of this flow — $\boxed{\text{1 point}}$
2. **[easy]** Sketch the location of the equilibria in the phase plane — $\boxed{\text{1 point}}$
3. Write down stability matrix **A** (in Strogatz, the "Jacobian") for this flow — $\boxed{\text{1 point}}$
4. Evaluate the stability matrix **A** for each equilibrium point — $\boxed{\text{2 point}}$
5. Compute the stability eigenvalues for each equilibrium point, indicate the kind (attracting node, hyperbolic, etc) — $\boxed{\text{3 point}}$
6. Indicate the direction of flow in the neighborhood of each equilibrium point — $\boxed{\text{3 point}}$
7. Indicate qualitative flow of all trajectories for $x < 0$ — $\boxed{\text{1 point}}$
8. **[more difficult]** Can this flow have a limit cycle (stable or unstable)? — $\boxed{\text{3 point}}$

Solution 3: Analysis of a 2-dimensional flow

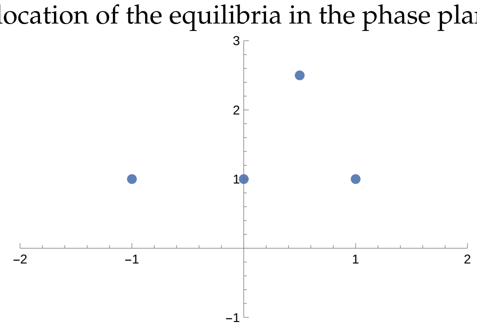
1. **[easy]** Zero flow-velocity solutions of (1)

$$0 = x(3 - y - 2x^2)\,, \qquad 0 = (y-1)(x - \frac{1}{2})\,. \tag{2}$$

are the equilibria of the flow:

$$(-1,1)\,,\ (\frac{1}{2},\frac{5}{2})\,,\ (1,1)\,,\ (0,1)\,. \tag{3}$$

2. **[easy]** Sketch the location of the equilibria in the phase plane



3. Write down stability matrix $\mathbf{A}$ for this flow

$$\mathbf{A} = \begin{bmatrix} 3 - y - 6x^2 & -x \\ y - 1 & x - \frac{1}{2} \end{bmatrix}. \tag{4}$$

4. Evaluate the stability matrix $\mathbf{A}$ for each equilibrium point

$$\begin{aligned} A_{(-1,1)} &= \begin{bmatrix} -4 & 1 \\ 0 & -\frac{3}{2} \end{bmatrix}, \qquad A_{(\frac{1}{2},\frac{5}{2})} = \begin{bmatrix} -1 & -\frac{1}{2} \\ \frac{3}{2} & 0 \end{bmatrix} \\ A_{(1,1)} &= \begin{bmatrix} -4 & -1 \\ 0 & \frac{1}{2} \end{bmatrix}, \qquad A_{(0,1)} = \begin{bmatrix} 2 & 0 \\ 0 & -\frac{1}{2} \end{bmatrix}. \end{aligned} \tag{5}$$

5. The eigenvalues are

$$\begin{aligned} (-1,1) &: \quad (\lambda_1, \lambda_2) = (-4, -\frac{3}{2}) \qquad \text{attracting node} \\ (\frac{1}{2}, \frac{5}{2}) &: \quad \mu \pm i\nu = \frac{1}{2}(-1 \pm i\sqrt{2}) \qquad \text{in spiral} \\ (1,1) &: \quad (\lambda_s, \lambda_u) = (-4, \frac{1}{2}) \qquad \text{hyperbolic} \\ (0,1) &: \quad (\lambda_u, \lambda_s) = (2, -\frac{1}{2}) \qquad \text{hyperbolic}. \end{aligned} \tag{6}$$

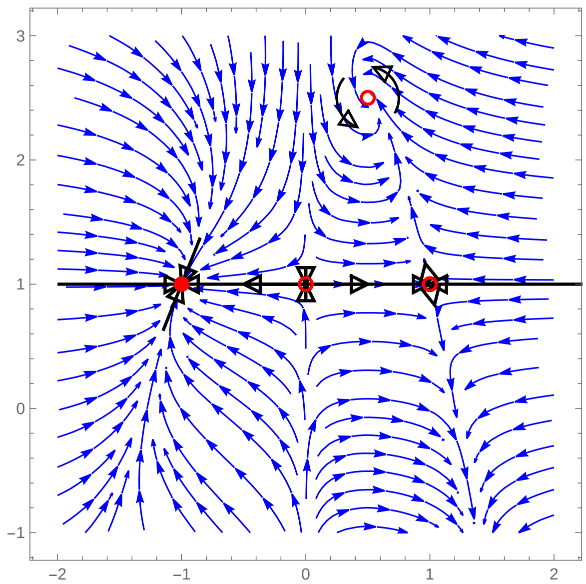
Shift the origin of (1) to the spiral attractor, $x \to x + 1/2$, $y \to y + 5/2$, take look close to the equilibrium:

$$\begin{aligned} \dot{x} &= (x + 1/2)\left(1/2 - 2(x + 1/2)^2 - y\right) \approx -(x + y/2) \\ \dot{y} &= (y + 3/2)x \approx 3x/2. \end{aligned} \tag{7}$$

By checking the direction of the flow $(\dot{x}, \dot{y})$ across nulclines $x = 0$ and $y = 0$ we determine that the in-spiralling is counter-clockwise. How fast is the in-spiralling? Angular period is $T = 2\pi/\nu_{(\frac{1}{2},\frac{5}{2})} = \pi\sqrt{2}/2$, and the radial contraction (in the linear neighborhood of $(\frac{1}{2}, \frac{5}{2})$ equilibrium) per period is

$$\Lambda = e^{\mu T} = e^{-\pi\sqrt{2}/2} =$$

6. Indicate the direction of flow in the neighborhood of each equilibrium point



Direction of the flow around the equilbria and a few trajectories, plotted in the (x, y) phase plane.

7. All trajectories initiated in the left half-plane $x < 0$ fall into the attracting node at $(-1, 1)$.

8. **[more difficult]** For 2-*d* flows trajectories either escape, fall into attracting points or attracting limit cycles. All trajectories initiated in the shifted lower right quadrant $x > 0, y < 1$ escape. So the only possibility of a limit cycle is in the upper right quadrant, an unstable cycle rotating counter-clockwise around the in-spiral $(\frac{1}{2}, \frac{5}{2})$. As the unstable manifold of $(1, 1)$ spirals into the $(\frac{1}{2}, \frac{5}{2})$, such unstable cycle is impossible.

Problem 4: A "2-d flow" with analytic stability exponent

One of the very few examples of flows for which the general solution $(q(t), p(t))$ can be evaluated analytically is the 2-d flow

$$\dot{q} = -\,p + q(1 - q^2 - p^2)\,, \qquad \dot{p} = q + p(1 - q^2 - p^2)\,. \tag{8}$$

You are asked to determine all periodic solutions of this flow, and determine analytically their stability exponents and multipliers.

1. Show that the 2-d flow (8) is separable in polar coordinates $q = r\cos\phi\,, p = r\sin\phi$: $\boxed{\text{1 point}}$

$$\dot{r} = r(1 - r^2)\,, \quad \dot{\phi} = 1\,. \tag{9}$$

2. What points/sets in state-space are invariant under this flow? $\boxed{\text{1 point}}$

3. Compute the stability matrix **A** for the $(q, p) = (0, 0)$ equilibrium from (8) and show that its eigenvalues are $\lambda \pm i\,\theta = 1 \pm i\,.$ $\boxed{\text{1 point}}$

4. Is the origin linearly stable or unstable? $\boxed{\text{1 point}}$

5. In the (r, ϕ) coordinates: Sketch the trajectory $x(t) = (q(t), p(t))$ of an initial point with $r < 1$ and an initial point with $r > 1$. Is the flow starting at any $r > 0$ attracted to or repelled from the $r = 1$? $\boxed{\text{1 point}}$

6. Show that after one period $T = 2\pi$ *for initial* $r \ll 1$, the radial stability multiplier per period $\Lambda = e^T$. $\boxed{\text{1 bonus}}$

7. Compute the stability matrix **A** for the $(q, p) \neq (0, 0)$ from (9), and show that it is diagonal in the (r, ϕ) coordinates, $\boxed{\text{1 point}}$

$$\mathbf{A} = \left[\begin{array}{cc} 1 - 3r^2 & 0 \\ 0 & 0 \end{array}\right]. \tag{10}$$

8. What invariance makes the $\lambda_\phi = 0$ eigenvalue vanish? $\boxed{\text{1 bonus}}$

9. Derive the radial stability multiplier $\Lambda_r = e^{-4\pi}$ per one period *in the neighborhood of* $r = 1$. Is this neighborhood very/weakly repelling/attracting? $\boxed{\text{2 bonus}}$

10. **[more difficult]** By multiplying (9) by r, separating variables and integrating $\boxed{\text{2 bonus}}$

$$\frac{r^2}{1 - r^2} = \frac{r_0^2}{1 - r_0^2} e^{2t}\,.$$

express the $r(r_0, t)$ trajectory in analytic form:

$$r(t)^2 = \frac{r_0^2}{r_0^2 + (1 - r_0^2)e^{-2t}}\,. \tag{11}$$

11. **[more difficult]** Show that the $[1\times 1]$ Jacobian matrix 2 bonus

$$J(r_0,t) = \left.\frac{\partial r(t)}{\partial r_0}\right|_{r_0=r(0)} . \tag{12}$$

satisfies

$$\frac{d}{dt}J(r,t) = A(r)\,J(r,t) = (1-3r(t)^2)\,J(r,t)\,, \qquad J(r_0,0)=1\,.$$

Integrate this by separating variables $d(\ln J(r,t)) = dt - 3r(t)^2 dt$, and show that the stability of a finite trajectory segment is:

$$J(r_0,t) = \frac{1}{(r_0^2+(1-r_0^2)e^{-2t})^{3/2}}e^{-2t}\,. \tag{13}$$

12. **[easy]** Does this general formula agree with the limit cycle contraction $\Lambda_r(1,t) = e^{-2t}$, and with the radial part of the equilibrium instability $\Lambda_r(r_0,t) = e^t$ computed above? 1 bonus

Solution 4: A "2-d flow" with analytic stability exponent

The 2-dimensional flow $x(t) = (q(t),p(t))$ is separable (check!) in polar coordinates $q = r\cos\phi\,,\quad p = r\sin\phi$:

$$\dot r = r(1-r^2)\,, \qquad \dot\phi = 1\,. \tag{14}$$

In the (r,ϕ) coordinates the flow starting at any $r>0$ is attracted to the $r=1$ limit cycle, with the angular coordinate ϕ wrapping around with a constant angular velocity $\nu = 1$. The non–wandering set of this flow consists of the $r=0$ equilibrium and the $r=1$ limit cycle.

equilibrium stability: As the change of coordinates is defined everywhere except at the equilibrium point ($r=0$, any ϕ), the equilibrium stability matrix has to be computed in the original (q,p) coordinates,

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}. \tag{15}$$

The eigenvalues are $\lambda = \mu \pm i\nu = 1 \pm i$, indicating that the origin is linearly unstable, with nearby trajectories spiralling out with the constant angular velocity $\nu = 1$. The Poincaré section ($p=0$, for example) return map is in this case also a stroboscopic map, strobed at the period (Poincaré section return time) $T = 2\pi/\nu = 2\pi$. The radial Floquet multiplier per one Poincaré return is $|\Lambda| = e^{\mu T} = e^{2\pi}$.

Limit cycle stability: From (14) the stability matrix is diagonal in the (r,ϕ) coordinates,

$$\mathbf{A} = \begin{bmatrix} 1-3r^2 & 0 \\ 0 & 0 \end{bmatrix}. \tag{16}$$

The vanishing of the angular $\lambda_\theta = 0$ eigenvalue is due to the rotational invariance of the equations of motion along ϕ direction. The expanding $\lambda_r = 1$ radial eigenvalue of the equilibrium $r = 0$ confirms the above equilibrium stability calculation. The contracting $\lambda_r = -2$ eigenvalue at $r = 1$ decreases the radial deviations from $r = 1$ with the radial Floquet multiplier $\Lambda_r = e^{\mu T} = e^{-4\pi}$ per one Poincaré return. This limit cycle is *very* attracting.

Stability of a trajectory segment: Multiply (14) by r to obtain $\frac{1}{2}\dot{r^2} = r^2 - r^4$, set $r^2 = 1/u$, separate variables $du/(1-u) = 2\,dt$, and integrate: $\ln(1-u) - \ln(1-u_0) = -2t$. Hence the $r(r_0, t)$ trajectory is

$$r(t)^{-2} = 1 + (r_0^{-2} - 1)e^{-2t} . \tag{17}$$

The $[1\times1]$ Jacobian matrix

$$J(r_0, t) = \left.\frac{\partial r(t)}{\partial r_0}\right|_{r_0 = r(0)} . \tag{18}$$

satisfies

$$\frac{d}{dt} J(r,t) = A(r)\, J(r,t) = (1 - 3r(t)^2)\, J(r,t) , \qquad J(r_0, 0) = 1 .$$

This too can be solved by separating variables $d(\ln J(r,t)) = dt - 3r(t)^2 dt$, substituting (17) and integrating. The stability of any finite trajectory segment is:

$$J(r_0, t) = (r_0^2 + (1 - r_0^2)e^{-2t})^{-3/2} e^{-2t} . \tag{19}$$

On the $r = 1$ limit cycle this agrees with the limit cycle multiplier $\Lambda_r(1,t) = e^{-2t}$, and with the radial part of the equilibrium instability $\Lambda_r(r_0, t) = e^t$ for $r_0 \ll 1$.